

RET: Respiratory Care Courses

Courses

RET 3028 Foundations of Respiratory Therapy

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

Prerequisite: BSC 1085/L AND BSC 1086/L AND CHM 2045/L AND MCB 2010/L

Co-requisite: RET 3028L

This lecture-based course offers a comprehensive introduction to the respiratory care profession. It provides theory and application of physics, chemistry, basic therapeutics, and disease management for respiratory therapists. This course focuses on the specific procedures and techniques used by respiratory care practitioners. It also covers infection control, therapeutic devices, patient assessment skills, medical gas administration, aerosol drug delivery, and an introduction to medical terminology using proper written and oral communication methods to prepare students for clinical practice.

RET 3028L Foundations of Respiratory Therapy Lab

College of Health, Department of Health Sciences & Admin

1 sh (may not be repeated for credit)

Prerequisite: BSC 1085/L AND BSC 1086/L AND CHM 2045/L AND MCB 2010/L

Co-requisite: RET 3028

This lab-based course reinforces understanding of the role of the respiratory care practitioner within the clinical setting. Emphasis is placed on correct setup and application of equipment, techniques, and therapies for respiratory care. Activities include medical gas administration, patient assessment skills, respiratory therapeutics, patient safety techniques, blood gas analysis, airway management equipment, lung expansion and bronchial hygiene devices, and humidification. Clinical skills and competencies will be demonstrated through live and virtual lab/simulation exercises.

RET 3266 Fundamentals of Mechanical Ventilation

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

Prerequisite: RET 3884

Co-requisite: RET 3266L

This lecture-based course provides an in-depth knowledge of theory, set-up, operation, and management of mechanical ventilators and related equipment. Principles and application of invasive and noninvasive ventilation will be explored. This course will cover the history and theory of mechanical ventilation, modes of ventilatory support, equipment implementation, patient monitoring techniques, ventilator management approaches, discontinuation strategies, and troubleshooting.

RET 3266L Fundamentals of Mechanical Ventilation Lab

College of Health, Department of Health Sciences & Admin

1 sh (may not be repeated for credit)

Prerequisite: RET 3884

Co-requisite: RET 3266

This lab-based course reinforces the principles, application, and management of mechanical ventilation by a respiratory care practitioner in a critical care setting. Emphasis is placed on cutting-edge ventilatory modes, noninvasive equipment, critical care monitoring technology, alternative patient management methods, and adjunctive therapies. Clinical skills and competencies will be demonstrated through live and virtual lab/simulation exercises.

RET 3354 Cardiopulmonary Pharmacotherapy

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

This lecture-based course is an introduction to an integrated, case-based evaluation of the physiology, pathophysiology, and pharmacology in the treatment and management of cardiopulmonary disorders. Students will study principles, practical uses, and interaction of pharmacological agents and their relationship to disease. Topic emphasis will be on using a whole-system's approach to patient care and the use of pharmacology in cardiopulmonary disease management.

RET 3445 Cardiopulmonary Diseases & Diagnostics

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

Prerequisite: RET 3884

Co-requisite: RET 3445L

This lecture-based course focuses on cardiopulmonary conditions, disease states, practices, and special procedures encountered in hospital, outpatient, and rehabilitation settings. Students will differentiate the etiology, clinical manifestations, pathogenesis, lab data, and treatment of common cardiopulmonary disorders. Emphasis is placed on diagnostics and therapeutic procedures for acute and chronic cardiopulmonary diseases. Problem-based learning will be used to develop comprehensive approaches to the practice of cardiorespiratory care.

RET 3445L Cardiopulmonary Diseases & Diagnostics Lab

College of Health, Department of Health Sciences & Admin

1 sh (may not be repeated for credit)

Prerequisite: RET 3884

Co-requisite: RET 3445

This lab-based course reinforces standard methodologies used to diagnose and monitor cardiopulmonary diseases. Students will develop foundational knowledge to identify indications, utilize equipment, and interpret findings of advanced cardiopulmonary diagnostic tests that assess different body systems. Emphasis is placed on technical aspects as well as disease presentation. Clinical competencies include pulmonary function and gas analysis equipment, polysomnography and sleep disorders, metabolic and exercise testing, ultrasound, bronchoscopy, and echocardiography. Clinical skills and competencies will be demonstrated through live and virtual lab/simulation exercises.

RET 3448 Advanced Therapeutics & Patient Monitoring Concepts in Respiratory Therapy

College of Health, Department of Health Sciences & Admin

2 sh (may not be repeated for credit)

Prerequisite: RET 3885

This lecture-based course reinforces advanced topics in clinical care expressed through a holistic approach to the monitoring of cardiopulmonary patients. Emphasis is placed on the etiology of complex illness and a systems-oriented approach to diagnosis, management, and treatment in the patient care setting. This course will cover monitoring modalities commonly utilized in respiratory care, analysis of lab values and their significance, various modalities of positive pressure ventilation as well as adjunct measurement tools that the respiratory therapist will encounter during clinical practice.

RET 3487 Cardiopulmonary Anatomy & Physiology

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

This lecture-based course focuses on cardiopulmonary anatomy and physiology essential to the practice of respiratory care. Emphasis is placed on normal and clinical physiological functions of the cardiovascular and pulmonary systems. Topics include cardiorespiratory functions, ventilatory mechanics, regulation of breathing, gas transport, acid-base balance, the cardiac cycle, hemodynamics, and diagnostic procedures used to evaluate these systems.

RET 3493 Patient Assessment

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

Prerequisite: BSC 1085/L AND BSC 1086/L AND CHM 2045/L AND MCB 2010/L

Co-requisite: RET 3493L

This lecture-based course will cover the fundamentals of cardiopulmonary assessment. This course introduces the role of the respiratory care practitioner in promoting a positive patient encounter. Learners will be introduced to skills that support a complete patient examination through the evaluation of medical records, physical findings, laboratory data, pulmonary function testing, imaging, hemodynamic monitoring, as well as diagnostic and therapeutic interventions.

RET 3493L Patient Assessment Lab

College of Health, Department of Health Sciences & Admin

1 sh (may not be repeated for credit)

Prerequisite: BSC 1085/L AND BSC 1086/L AND CHM 2045/L AND MCB 2010/L

Co-requisite: RET 3493

This lab-based course reinforces the role of the respiratory care practitioner throughout a cardiopulmonary assessment. Emphasis is placed on correct clinical skills that support a complete patient examination. Activities include the assessment of medical records, physical findings, laboratory data, pulmonary function tests, medical images, hemodynamic monitoring data, as well as the evaluation of diagnostic and therapeutic interventions. Clinical skills and competencies will be demonstrated through live and virtual lab/simulation exercises.

RET 3884 Clinical Practicum I

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

Prerequisite: BSC 1085/L AND BSC 1086/L AND CHM 2045/L AND MCB 2010/L

This clinical practice course focuses on the application of basic therapeutic techniques and procedures. Clinical practice courses require the student to integrate theory and lab/simulation training in the patient care setting. This course provides supervised entry-level clinical experience via assigned rotations at medical facilities. Clinical competencies include clinical documentation, patient assessment, patient safety techniques, respiratory therapeutics and diagnostics, blood gas analysis, medical gas and medication administration, and airway care within the scope of practice for respiratory care. Students will demonstrate, through performance evaluations, clinical competence and novice proficiency of patient care in a clinical setting.

RET 3885 Clinical Practicum II

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

Prerequisite: RET 3884

This clinical practice course further develops both basic and advanced skills required in the intensive care of the cardiopulmonary patient. Clinical practice courses require the student to integrate theory and lab/simulation training in the patient care setting. This course provides supervised clinical experience in the critical care units via assigned rotations at medical facilities. Topics include continuous Clinical Practicum I duties, airway care skills, mechanical ventilation management, patient stabilization techniques, invasive and noninvasive monitoring, hemodynamic evaluations, and cardiopulmonary diagnostics within the scope of practice for respiratory care. Students are also given the opportunity to begin developing their neonatal-pediatric critical care skills. Students will demonstrate, through performance evaluations, clinical competence and appropriate levels of critical care knowledge in a clinical setting.

RET 4050 Evidence-Based Practice in Respiratory Care

College of Health, Department of Health Sciences & Admin

2 sh (may not be repeated for credit)

Prerequisite: HSC 4050

This lecture-based course focuses on the concept of evidence-based practice and the role research plays in the field of respiratory care. Students will acquire the skills necessary to incorporate evidence and best practices into their professional work. This course will cover research methods, hierarchy of evidence, critical appraisal skills, and quality improvement initiatives. Each student will further develop their plan to collect and present data for the respiratory therapy capstone project.

RET 4277 Critical Care Management

College of Health, Department of Health Sciences & Admin

3 sh (may not be repeated for credit)

Prerequisite: RET 3885

Co-requisite: RET 4277L

This lecture-based course focuses on the theory and clinical application of adult critical care management techniques. Emphasis is placed on advanced methods of information gathering and decision making for the respiratory care practitioner. Topics include disease management in the intensive care unit, invasive and non-invasive patient monitoring, hemodynamic assessment, pharmacological interventions, and special procedures.

RET 4277L Critical Care Management Lab

College of Health, Department of Health Sciences & Admin

1 sh (may not be repeated for credit)

Prerequisite: RET 3885

Co-requisite: RET 4277

This lab-based course reinforces techniques, procedures, and management strategies used to support the critically ill adult patient. Emphasis is placed on correct setup and application of equipment, techniques, and therapies for adult critical care. Clinical competencies include intubation and airway care, invasive and noninvasive support modalities, hemodynamic monitoring tools, pharmacological interventions, and other specialty diagnostic procedures. Clinical skills and competencies will be demonstrated through live and virtual lab/simulation exercises.

RET 4535 Disease Management: Extended Respiratory Therapy Services

College of Health, Department of Health Sciences & Admin

2 sh (may not be repeated for credit)

Prerequisite: RET 4886

This lecture-based course focuses on the evolution and trends of extended respiratory therapy services. This course provides an in-depth study of cardiopulmonary disease management and alternatives to hospital care. Emphasis is placed on the principles and practices applied to extended care environments. This course will cover virtual care services, outpatient cardiopulmonary rehabilitation, interventional pulmonology, DME services, pulmonary diagnostics, palliative care, patient and family education, smoking cessation programs, home health, and sub-acute care facilities.

RET 4616 Professional Healthcare Presence: Leadership, Administration, & Education

College of Health, Department of Health Sciences & Admin

2 sh (may not be repeated for credit)

Prerequisite: RET 3885

This lecture-based course explores leadership qualities, administrative skills, and educational techniques appropriate to the advancement of the respiratory therapist. It introduces students to leadership theories and perspectives in healthcare and provides a foundation for future healthcare leaders. Additionally, students will develop an understanding of the administrative and financial aspects of healthcare operations. Emphasis is also placed on students applying curricular theory to best practices in the classroom, disease care, and patient education settings.

RET 4718 Neonatal-Pediatric Respiratory Care

College of Health, Department of Health Sciences & Admin

2 sh (may not be repeated for credit)

Prerequisite: RET 3885

Co-requisite: RET 4718L

This lecture-based course focuses on the theoretical application of clinical care specific to neonatal and pediatric patients. Students will utilize evidence-based knowledge and critical thinking skills in the comprehensive respiratory care of newborns, infants, children, and adolescents along the illness-wellness continuum. This course will cover fetal growth and development, neonatal and pediatric cardiopulmonary physiology and pathophysiology, respiratory assessment of the neonatal and pediatric patients, as well as respiratory care diagnostic and therapeutic interventions targeted at specific cardiopulmonary pathologies.

RET 4718L Neonatal-Pediatric Respiratory Care Lab

College of Health, Department of Health Sciences & Admin

1 sh (may not be repeated for credit)

Prerequisite: RET 3885

Co-requisite: RET 4718

This lab-based course reinforces techniques, procedures, and management strategies specific to neonatal and pediatric patients. Emphasis is placed on correct setup and application of equipment, techniques, and therapies for neonates and pediatric patient populations. Clinical competencies include neonatal-pediatric patient assessments, resuscitation methods, intubation and airway care, invasive and noninvasive support modalities, hemodynamic monitoring tools, pharmacological interventions, and other specialty diagnostic procedures. Clinical skills and competencies will be demonstrated through live and virtual lab/simulation exercises.

RET 4886 Clinical Practicum III

College of Health, Department of Health Sciences & Admin

4 sh (may not be repeated for credit)

Prerequisite: RET 3885

This clinical practice course provides an opportunity for students to acquire respiratory care experience with neonatal and pediatric patients in the healthcare setting. Clinical practice courses require the student to integrate theory and lab/simulation training in the patient care setting. This course provides supervised clinical experience in the neonatal and pediatric areas of medical facilities via assigned rotations. Clinical competencies include continuous Clinical Practicum I and II duties, as well as neonatal-pediatric patient assessments, labor and delivery assistance, resuscitation methods, pharmacological interventions, and other specialty diagnostic procedures within the scope of practice for respiratory care. Students are also given the opportunity to further develop their adult critical care skills. Students will demonstrate, through performance evaluations, clinical competence and appropriate levels of critical care knowledge in a clinical setting.

RET 4887 Clinical Practicum IV

College of Health, Department of Health Sciences & Admin

4 sh (may not be repeated for credit)

Prerequisite: RET 4886

This clinical practice course provides an opportunity for students to advance their respiratory care expertise with adult, pediatric, and neonatal patient populations. Clinical practice courses require the student to integrate theory and lab/simulation training in the patient care setting. This course provides supervised clinical experience on the critical care units of acute care hospitals. Extended care settings such as (but not limited to) sub-acute care, sleep, home health, and pulmonary rehabilitation will also be integrated into this clinical experience. All clinical competencies expected of new graduates for entry into respiratory therapy practice must be completed. Students will demonstrate, through performance evaluations, clinical competence and professional levels of all respiratory care-related knowledge and skills in a clinical setting.

RET 4930 RT Seminar: Exam Preparation & Career Guidance

College of Health, Department of Health Sciences & Admin

1 sh (may not be repeated for credit)

Prerequisite: RET 4886

This workshop-style course provides preparation and reflection on the scope of practice for respiratory therapy students entering the profession. Students will review respiratory care across the lifespan while highlighting problem-solving and decision-making skills. It gives each student an assessment of their educational strengths and weaknesses with respect to the practice of respiratory care. Emphasis is placed on competencies listed in the content matrices for the National Board for Respiratory Care (NBRC) credentialing exams.

RET 4950 Respiratory Therapy Capstone Project

College of Health, Department of Health Sciences & Admin

2 sh (may not be repeated for credit)

Prerequisite: HSC 4050 AND RET 4050

This project-focused course includes experiential-learning in a chosen focus area (clinical, education, administrative, or population-based). This experience will culminate in a respiratory therapy-related capstone project in the form of research, or other scholarly activity that articulates the design, organization, implementation, data analysis, interpretation of results, and use of research to support evidence-based practice. Students will apply knowledge and skills learned in the program as well as utilize clinical expertise and best evidence to complete their projects. Successful completion of this course includes both an oral and written presentation of the project.